



CHANDIGARH UNIVERSITY

Discover. Learn. Empower.

ASSIGNMENT-1

Subject: Computer Networks

Subject Code: 24CST-339

Submitted by:

Name: Mehak Mehra

UID: 24BAI70075

Branch/ Section: 24AIT_KRG-1_B

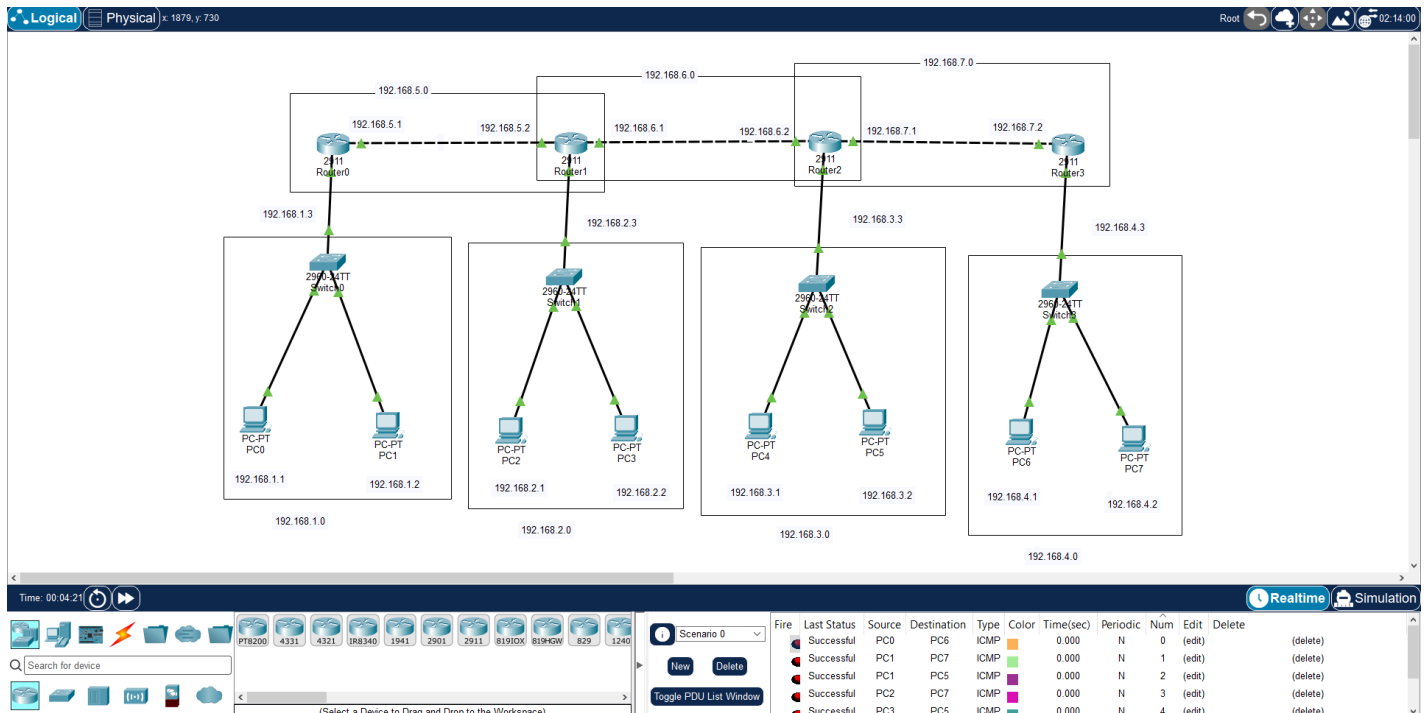
Submitted to:

Faculty Name: Alok Sir

Assignment – 1

Q1. Develop Static Routing with 4 PC's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be Written in text format as well. (6 M)

Ans.



The screenshot shows a fully configured 4-router static network where all subnets are highlighted with their Network IDs and individual IP addresses, with end-to-end connectivity successfully tested across all devices.

Q2. What is OSI model? What are the different layers of OSI model? Write the functionality, Data format and Protocols used at each layer. (3 M)

Ans. The OSI (Open Systems Interconnection) model is a 7-layer reference model developed by ISO to standardise how data is transmitted between devices over a network. Each layer performs a specific function and communicates with the layer directly above and below it.

Layer	Functionality	Data Format	Protocols
-------	---------------	-------------	-----------

7. Application	Provides network services directly to end users/applications such as web, email and file transfer.	Data/Message	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Translates data formats, encryption/decryption and compression.	Data	SSL/TLS, JPEG, MPEG, ASCII
5. Session	Establishes, manages and terminates communication sessions between applications.	Data	NetBIOS, RPC, PPTP
4. Transport	Provides end-to-end delivery, reliability, flow control, error recovery and segmentation.	Segment (TCP) / Datagram (UDP)	TCP, UDP
3. Network	Determines the best path and handles logical addressing and routing between networks.	Packet	IP, ICMP, OSPF, BGP
2. Data Link	Provides node-to-node delivery, framing, MAC addressing and error detection.	Frame	Ethernet, Wi-Fi (802.11), PPP
1. Physical	Transmits raw bits through the physical medium using electrical/radio/optical signals.	Bits	Ethernet PHY, USB, Bluetooth PHY

Q3. What are the different Error Detection methods? Explain each with the help of proper examples. (3 M)

Ans. Error detection is the process of detecting whether the data received by the receiver has been changed or corrupted during transmission.

The main error detection methods are:

1. Parity Check
2. Checksum
3. Cyclic Redundancy Check (CRC)

1. PARITY CHECK

A parity bit is an extra bit added to the data to make the number of 1s either even (even parity) or odd (odd parity).

Example: Even Parity

Suppose the data is: 1011001

Count the number of 1s: 4 ones

There are already 4 ones, which is even.

Therefore, for even parity, we add: Parity bit = 0

Transmitted data: 10110010

Now suppose during transmission one bit changes:

10100010

The receiver counts the 1s: 3 ones

There are 3 ones, which is odd.

Since even parity was expected, the receiver detects that an error has occurred.

2. CHECKSUM

In the checksum method, the sender divides the data into fixed-size blocks, adds them using one's complement addition, and takes the one's complement of the sum. This value is called the checksum.

The checksum is sent along with the original data.

Example:

Suppose we have two 4-bit data blocks:

Data 1 = 1010

Data 2 = 1100

Step 1: Add the blocks

$1010 + 1100 = 10110$

We have a 5-bit result, so the leftmost carry is added back to the remaining 4 bits:

$0110 + 1 = 0111$

Step 2: Take one's complement

Change every 0 to 1 and every 1 to 0:

Sum = 0111

Checksum = 1000

So the sender transmits: 1010 1100 1000 → the data and the checksum.

The receiver adds all three: $1010 + 1100 + 1000 = 11110$

We have a 5-bit result, so the leftmost carry is added back to the remaining 4 bits:

$1110 + 1 = 1111$

Since the result is all 1s, we know that there is no error.

3. CYCLIC REDUNDANCY CHECK (CRC)

CRC is a powerful error detection technique that uses binary division.

The sender and receiver agree on a generator/divisor. The sender divides the data by this generator using XOR division. The remainder is attached to the original data.

Example

Data = 1101

Divisor = 1011

The divisor has 4 bits, so we append $4 - 1 = 3$ zeros to the data:

Data = 1101

After appending 3 zeros = 1101000

Now perform binary division using XOR.

$1101000 \div 1011$

Continuing the XOR division gives a remainder: 001

Therefore, CRC remainder = 001

The sender attaches this remainder to the original data:

Original data = 1101

CRC = 001

Transmitted data = 1101001

At the receiver

The receiver divides:

$1101001 \div 1011$

If the remainder is: 000, then the receiver assumes that no error has been detected.

If the remainder is non-zero, then the receiver knows that the data was corrupted during transmission.